

FROM ORDINARY TO BAYESIAN SHRINKAGE ESTIMATORS

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SUMMARY

In the first part of this paper the problem of shrinking the minimum variance unbiased linear estimator $\hat{\theta}$ of the parameter θ towards a natural origin θ_0 is studied for various distributions. An estimator $T_L = k\hat{\theta} + (1-k)\theta_0$ with k a constant, $0 \leq k \leq 1$, is compared with estimators proposed by Mehta and Srinivasan (1971), Thompson (1968) and Pandey (1979). The relationship between the different estimators is examined. The estimator T_L is shown to be better than the other estimators in the sense of having a smaller mean squared error in an interval around θ_0 and is moreover the simplest for computational and interpretation purposes. Secondly, attention is paid to the question whether it would perhaps be more realistic to postulate a prior distribution for θ around θ_0 and use an ordinary Bayes estimator, $\hat{\theta}_0$, instead of $\hat{\theta}$ in T_L . The approach is very general and a wide variety of cases are studied.

1. Introduction

We consider the problem of estimating the parameter θ of a distribution when information regarding θ is available in the form of a prior distribution of θ around θ_0 . The general case is considered where the prior distribution places a weight $(1-b)$ in the value θ_0 and distributes the rest of the probability mass b , $0 \leq b \leq 1$, over some suitable interval. If $\hat{\theta}_0$ denotes the Bayes estimator of θ based on this prior distribution, we consider an estimator

$$(1.1) \quad T_B = k\hat{\theta} + (1-k)\hat{\theta}_0, \quad 0 \leq k \leq 1.$$

By varying k and b , a broad spectrum of cases are obtained. If $k=0$ and $b=1$, T_B becomes an ordinary Bayes estimator. If $b=0$ T_B reduces to the ordinary shrinkage estimator

$$(1.2) \quad T_L = k\hat{\theta} + (1-k)\theta_0.$$

Besides studying the properties of T_B for different values of k and b , comparisons are also made with other estimators proposed in the literature. In order to avoid confusion, we firstly study the ordinary shrinkage estimator T_L and its relationship with other shrinkage estimators, and thereafter we return to T_B .

2. Unified theory for Pandey-type shrinkage estimators

Consider a distribution with unknown parameter θ and let $\hat{\theta}$ be an unbiased estimator of θ . (Usually $\hat{\theta}$ is a MVULE of θ , but not necessarily.) As in Pandey (1979) we consider the estimator

$$(2.1) \quad T = a[k\hat{\theta} + (1-k)\theta_0], \quad 0 \leq k \leq 1$$

with a some constant to be specified later and k a constant specified by the experimenter according to his belief $(1-k)$ in θ_0 .

We present a unified discussion of four apparently closely related estimators. We do this by looking at the mean squared error of the estimators, i.e. for T we have

$$(2.2) \quad \begin{aligned} \text{MSE}(T) &= E(T - \theta)^2 \\ &= a^2 k^2 \text{var}(\hat{\theta}) + \theta^2 (ad - 1)^2 \end{aligned}$$

with $d = k + (1-k)\rho$ and $\rho = \theta_0/\theta$.

The relative efficiency of the estimator T with respect to $\hat{\theta}$ is

$$(2.3) \quad \begin{aligned} \text{REF}(T, \hat{\theta}) &= \text{MSE}(\hat{\theta})/\text{MSE}(T) \\ &= \text{var}(\hat{\theta})/(a^2 k^2 \text{var}(\hat{\theta}) + \theta^2 (ad - 1)^2) \\ &= [a^2 k^2 + \theta^2 (ad - 1)^2 / \text{var}(\hat{\theta})]^{-1} \\ &= [a^2 k^2 + (ad - 1)^2 \delta^2]^{-1} \text{ where} \end{aligned}$$

$$(2.4) \quad \delta = \frac{1}{\theta / (\text{var}(\hat{\theta}))^{1/2}}.$$

Clearly the estimator for which $\text{MSE}(T)$ is a minimum, is the best. Thompson (1968) considered the special case $a = 1$ and determined the best value of k by solving for k in $\partial \text{MSE}(T) / \partial k = 0$. His estimator is

$$(2.5) \quad T_T = (\hat{\theta} - \theta_0)^3 / [(\hat{\theta} - \theta_0)^2 + h \text{var}(\hat{\theta})] + \theta_0$$

with $\text{var}(\hat{\theta})$ the estimated variance of $\hat{\theta}$ and $h = 1$. The constant h has subsequently been introduced by Thompson in order to make his estimator "more flexible". T_T is clearly not a linear combination between $\hat{\theta}$ and θ_0 . The method proposed by Pandey (1979) is to consider k as a constant and to solve for a in $\partial \text{MSE}(T) / \partial a = 0$. We obtain

$$\begin{aligned} a &= d\theta^2/(k^2\text{var}(\hat{\theta}) + d^2\theta^2) \\ &= d^2\delta^2/(k^2 + d^2\delta^2) \end{aligned}$$

and hence

$$(2.6) \quad T = d\theta^2[k\hat{\theta} + (1-k)\theta_0]/(d^2\theta^2 + k^2\text{var}(\hat{\theta})).$$

Unfortunately the estimator T is of no practical value because it involves the unknown parameter θ . According to Pandey we now estimate θ by $\hat{\theta}$ in (2.6) and obtain

$$(2.7) \quad T_P = \hat{d}^2\hat{\theta}^3/(\hat{d}^2\hat{\theta}^2 + h k^2\text{var}(\hat{\theta}))$$

with $\hat{d} = k + (1-k)\hat{\rho}$, $\hat{\rho} = \theta_0/\hat{\theta}$ and $h = 1$. We again introduce the constant h to make the estimator more flexible. The relative efficiency of T_P with respect to $\hat{\theta}$ can be determined from $\text{MSE}(T_P)$, which naturally differs from $\text{MSE}(\hat{\theta})$.

On comparing $\text{REF}(T, \hat{\theta})$ and $\text{REF}(T_P, \hat{\theta})$ in estimating the variance of a normal distribution (Pandey 1979), it appears that T_P is inferior to T . This is obviously due to the fact that T_P depends more heavily on $\hat{\theta}$ than T . In a similar context Efron and Morris (1973) said: "It is obvious that a price must be paid for estimating B rather than knowing its true value."

From (2.3)

$$(2.8) \quad \text{REF}(T, \hat{\theta}) = [a^2k^2 + (ak - 1 + \rho a(1-k))^2\delta^2]^{-1} \text{ and}$$

we can try to maximize this expression by giving a and k values which are not based on $\hat{\theta}$. Since δ is generally an increasing function of n , it seems important to make the coefficient of δ^2 , i.e. $ak - 1 + \rho a(1-k)$, equal to zero. Hence $a = 1/(k + \rho(1-k))$. We do not know the value of ρ , but if our guess θ_0 was close to θ , ρ would be close to 1. For $\rho = 1$ we get $a = 1$, which indicates that our ultimate choice of a must be close to 1. Taking $a = 1$ in (2.1) leads to the estimator proposed in Lemmer (1981) namely

$$(2.9) \quad T_L = k\hat{\theta} + (1-k)\theta_0, 0 \leq k \leq 1$$

with $(1-k)$ directly proportional to the experimenter's confidence in θ_0 .

The formal relationship between T_T , T_P and T_L is now clear. It can furthermore be seen that T_L is a special case of the Mehta and Srinivasan (1971) estimator

$$(2.10) \quad T_M = \hat{\theta} - a'(\hat{\theta} - \theta_0)\exp[-b'(\hat{\theta} - \theta_0)^2/\text{var}(\hat{\theta})]$$

with a' and b' constants, because if $b' = 0$ and $a' = (1 - k)$, T_M reduces to T_L . This immediately gives meaning to the constant a' (Mehta and Srinivasan did not interpret a' as the experimenter's belief in θ_0). We appreciate their motivation for the introduction of the exponential term (to bound $\text{REF}(T, \hat{\theta})$), but we shall show why we prefer to use T_L rather than T_M .

3. Comparison of estimators

3.1 Introduction

Following Mehta and Srinivasan (1971) we define the *effective interval* of any estimator T as the interval around θ_0 where $\text{REF} \equiv \text{REF}(T, \hat{\theta})$ is greater than one, i.e. where T is better than $\hat{\theta}$. In our comparison of estimators we are guided by two considerations namely the length of the effective interval of T and the maximum value of REF , but contrary to Mehta and Srinivasan we do not require REF to be bounded from below. Our philosophy is that we weigh our confidence in the guessed value θ_0 by the factor $(1 - k)$: If we do not have much confidence in our guess θ_0 , we take k large (say ≥ 0.7), with the result that the effective interval is long, thus giving a reasonable protection against a wrongly guessed value — cf. Figures 1 and 2. If, on the other hand, we have a fairly good idea of the value of θ_0 , by taking k small we have REF very large for θ in the vicinity of θ_0 , which means that T_L is a much better estimator than $\hat{\theta}$. Retention of the constant b in T_M has the effect of bending the REF curve upward at some point far away from θ_0 at the expense of shortening the effective interval. We consider this as a great disadvantage because outside the effective interval we do not see much sense in having REF at θ_2 larger than at θ_1 for $\theta_0 < \theta_1 < \theta_2$ — we therefore prefer T_L to T_M . For numerical comparisons of the different estimators we consider estimation of the mean of the normal, Poisson and Gamma distributions. A comparison of these estimators for the parameter π of the binomial (n, π) distribution has been presented in Lemmer (1981).

3.2 Estimation of the mean in a normal (μ, σ^2) distribution

We consider only the case where σ^2 is unknown (for σ^2 known the results are very similar) and denote the sample mean and variance, based on n observations, by $\bar{x}$ and s^2 respectively. Our estimator is

$$(3.1) \quad T_L = k\bar{x} + (1 - k)\mu_0$$

and that of Mehta and Srinivasan (1971)

$$(3.2) \quad T_M = \bar{x} - a'(\bar{x} - \mu_0) \exp[-nb'(\bar{x} - \mu_0)^2/s^2].$$

Due to the invariance of the normal distribution under translation, we may for the purposes of our study take $\mu_0 = 0$ without any loss of generality. The efficiency of T_P in this case is the same as that of T_T , which can be seen if we take $\mu_0 = 0$ in

$$(3.3) \quad T_T = (\bar{x} - \mu_0)^3 / [(\bar{x} - \mu_0)^2 + h s^2/n] + \mu_0$$

and

$$(3.4) \quad T_P = \hat{d}^2 \bar{x}^3 / (\hat{d}^2 \bar{x}^2 + h k^2 s^2/n)$$

$$\text{with } \hat{d} = k + (1-k)\mu_0/\bar{x}.$$

$MSE(T_T)$ and $MSE(T_M)$ are calculated by Romberg quadrature where

$$(3.5) \quad MSE(T_T) = \frac{\sqrt{n}}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (T_T - \mu)^2 \exp[-n(\bar{x} - \mu)^2/2\sigma^2] d\bar{x}$$

and $MSE(\bar{x}) = \sigma^2/n$. For T_L we have, from (2.8), $REF = [k^2 + (1-k)^2\delta^2]^{-1}$ and with $\delta = (\mu - \mu_0)\sqrt{n}/\sigma = \mu\sqrt{n}/\sigma$. For a comparison of the different estimators for $n=9$ we considered the following cases: T_M with $a' = 0,302$; $b' = 0,01$ and $a' = 0,379$; $b' = 0,01$ (cf. Mehta and Srinivasan, 1971 Fig. 2), T_T with $h = 1$ and 10 , T_L with $k = 0,25$; $0,50$; $0,71$ and $0,75$ - cf. Figure 1. A comparison of T_L , $k = 0,25$ and T_T , $h = 10$ shows that the effective interval of T_L is longer than that of T_T and within this interval T_L is everywhere more effective than T_T . By taking $k = 0,5$ the effective interval of T_L is markedly longer than for $k = 0,25$, but REF is smaller at $\delta = 0$, although still much greater than 1. T_M with $a' = 0,302$; $b' = 0,01$ and T_L with $k = 0,71$ have the same value of REF at $\delta = 0$ and are thus easily comparable. The effective interval of T_L is longer than that of T_M . The small value of b' has the effect that the REF -curves of T_M and T_L are very similar, with the former reaching a minimum and increasing to 1 as δ increases. As we are mainly interested in the performance of the estimators for δ close to 0, we prefer T_L to T_M .

3.3 Mean of a Gamma (μ) distribution

Consider a random sample $X_1, \dots, X_n$ from a gamma (μ) distribution, then $U = n\bar{X}$ has a gamma (η) distribution where $\eta = n\mu$. The different estimators for μ are:

$$(3.6) \quad T_P = [k\bar{x} + (1-k)\mu_0]^2 \bar{x} / \{[k\bar{x} + (1-k)\mu_0]^2 + h k^2 \bar{x}/n\}$$

$$(3.7) \quad T_T = [(n\bar{x} - \eta_0)^3 / \{(n\bar{x} - \eta_0)^2 + h n \bar{x}\} + \eta_0] / n \text{ with } \eta_0 = n\mu_0$$

$$(3.8) \quad T_M = \bar{x} \{1 - a' \exp[-b' (n\bar{x} - \eta_0)^2 / n\bar{x}] + a' \mu_0 \exp[-b' (n\bar{x} - \eta_0)^2 / n\bar{x}]\}$$

and

$$(3.9) \quad T_L = k\bar{x} + (1 - k)\mu_0.$$

T_P no longer behaves similar to T_T and thus has to be considered separately. For an estimator T of μ we have

$$\text{MSE}(T) = \int_0^\infty (T - \mu)^2 x^{\eta-1} e^{-x} dx / \Gamma(\eta)$$

with $\eta = n\mu$. Hence the MSE of T_P , T_T and T_M can be calculated by quadrature, while $\text{MSE}(\bar{x}) = \eta/n^2$. For T_L we have $\text{REF} = [k^2 + [(1-k)(\eta_0 - \eta)]^2 / \eta]^{-1}$.

Despite using a wide variety of values for h and k , the Pandey-type estimator T_P is very bad compared to the other estimators considered, and is therefore not presented in the figure. In Figure 2 we have a comparison of estimators for $\eta_0 = 10$ and the following cases: T_M with $a' = 0,329$; $b' = 0,001$, T_T with $h = 1$ and 10 , T_L with $k = 0,25$; $0,50$; $0,672$ (to be compared with T_M) and $0,75$. Over the range of η values considered, the REF curves of T_M and T_L ($k = 0,672$) coincide, as we could expect from the small value of $b' = 0,001$. T_T with $h = 10$ has a shorter effective interval than T_L with $k = 0,25$, and T_T with $h = 1$ is clearly not good compared to T_L with $k = 0,672$ and T_M . Similar results have been obtained for the case $\eta_0 = 6$. Our results confirm those of Mehta and Srinivasan (1971), namely that T_M is better than T_T .

3.4 Mean of a Poisson (λ) distribution

Consider a random sample $X_1, \dots, X_n$ from a Poisson distribution with parameter λ . The estimators for λ are the same as in (3.6) – (3.9) with μ_0 replaced by λ_0 . The MSE for T_P , T_T and T_M are calculated from

$$(3.10) \quad \begin{aligned} \text{MSE}(T) &= \sum_{r=0}^{\infty} (T - \lambda)^2 e^{-n\lambda} (n\lambda)^r / r! \\ &= \sum_{r=0}^{\infty} (T - \eta/n)^2 e^{-\eta} \eta^r / r! \quad \text{with } \eta = n\lambda \end{aligned}$$

and $n\bar{x} = r$. Furthermore $\text{MSE}(\bar{x}) = \eta/n^2$. For T_L we again have

$$\text{REF} = [k^2 + [(1 - k)(\eta_0 - \eta)]^2 / \eta]^{-1}.$$

The different estimators have been compared for $\eta_0 = 6$ and 10 . The figures (not presented here) resemble those of the gamma distribution. The conclusions are similar to those in the

case of the gamma and normal distributions, and also the binomial distribution presented in Lemmer (1981).

3.5 Conclusions

The estimator T_L , which can be considered as a special case of the Mehta and Srinivasan (1971) estimator T_M , gives new meaning to the constant a' in T_M . The experimenter who prefers to use T_M due to its property of having a bounded REF curve, has a rational way of choosing the value of a' , namely in direct proportion to his confidence in the guessed value θ_0 . We nevertheless prefer T_L to T_M because over the interval of main interest it is more effective than T_M , and it is much easier to compute. If $(1 - k)$ is large, T_L is a very effective estimator, and when $(1 - k)$ is moderate to small, T_L remains fairly good over a long effective interval.

4. Bayesian shrinkage estimators

4.1 Introduction

After studying ordinary shrinkage estimators such as T_L , it is only natural to question the experimenter's ability to express his prior knowledge of θ by a single value θ_0 . It seems more natural to specify a *distribution* for θ around θ_0 , possibly with a weight $(1 - b)$ in the value θ_0 , as explained in section 1. We thus return to the estimator $T_B = k\theta + (1 - k)\theta_0$ with θ_0 the Bayes estimator derived from a prior distribution which places a weight $(1 - b)$ in θ_0 and distributes the rest of the probability mass b according to some probability distribution. Our main interest is in examining the effect of varying the values of k and b — the choice of the prior distribution is of secondary importance. We therefore found it convenient for control purposes to use the same prior distributions as Mood and Graybill (1963).

The technique of combining a Bayes and ordinary estimator and analyzing the estimator's properties by the frequency approach is not new — cf. the discussion of empirical Bayes estimators by Efron and Morris (1973) par. 2 and 9. We shall not go further into the philosophy of this approach.

For illustration purposes we consider estimators of the mean of the normal and Poisson distributions, and of the parameter π of the binomial distribution.

4.2 The Binomial distribution

Let $X_1, \dots, X_n$ be independent Bernoulli (π) random variables where the parameter π has the a priori distribution

$$f(\pi) = \begin{cases} b & \text{for } 0 \leq \pi \leq 1, \pi \neq \pi_0 \\ 1-b & \text{for } \pi = \pi_0 \end{cases}$$

which places a weight $1 - b$ in the point π_0 and distributes the remaining mass b uniformly over the interval $[0, 1]$.

The Bayes estimator for π , based on this prior distribution, is given by

$$\hat{\pi}_0 = \frac{(1-b)\pi_0^{r+1}(1-\pi_0)^{n-r} + bB(r+2, n-r+1)}{(1-b)\pi_0^r(1-\pi_0)^{n-r} + bB(r+1, n-r+1)}$$

with $B(\cdot, \cdot)$ the ordinary beta function and $r = \sum_{i=1}^n x_i$, the number of successes.

If $b = 1$ $\hat{\pi}_0$ is the ordinary Bayes estimator:

$$(r+1)/(n+2).$$

Our Bayesian shrinkage estimator is $T_B = kp + (1-k)\hat{\pi}_0$ with $p = r/n$. For comparison purposes we calculate the relative efficiency $REF(T_B, p) = MSE(p) / MSE(T_B)$ where

$$MSE(T_B) = \sum_{r=0}^n (T_B - \pi)^2 \binom{n}{r} \pi^r (1-\pi)^{n-r}$$

and $MSE(p) = \pi(1-\pi)/n$.

$REF(T_B, p)$ has been calculated for a variety of parameter values. For $n = 8$, $\pi_0 = 0,5$ we present in figure 3 the relative efficiencies for $\pi \geq 0,5$ (in this case all curves are symmetric about 0,5). Curves 1 and 3 reach maximum values 9,4 and 16 respectively. If k and b are small, REF is high around $\pi = \pi_0$, but the effective interval is shorter than when k and b are larger. Similar results have been obtained in the cases $\pi_0 = 0,3$ and $\pi_0 = 0,1$ - cf. figures 2 and 3 of Lemmer (1980). Amongst the cases considered the estimator with $k = 0,25$ and $b = 0$ (curve 3) is obviously the best over a fairly long interval. In general both k and b should hopefully be small if we desire a high REF around $\pi = \pi_0$. Specifically, b rather than k should be small: Case 3: $k = 0,25$; $b = 0$ is better than case 1: $k = 0$; $b = 0,25$ in the sense of a higher REF around π_0 and a longer effective interval. If we compare curves 2, 4 and 6 we notice that the latter is not quite as high in $\pi = \pi_0$ as curves 2 and 4, but has the longest effective interval in all 3 cases. Thus curve 6 is generally preferable to 2 and 4. In cases 5: $k = 0,25$; $b = 0,5$ and 7: $k = 0,5$; $b = 0,25$ REF is never very large, but the effective interval is fairly long and REF does not decline as far below 1 as in the other cases considered. The ordinary Bayes estimator ($k = 0$; $b = 1$) has a very long effective interval, but at its best it is not much better than the ordinary estimator p .

If we are very confident about our guess of π_0 , case 3 is obviously the best value combination, but the very high REF around π_0 is not so important - in the case of a little less confidence case 6 may be a better combination: It still has a fairly high REF around π_0 and a reasonably longer effective interval. Note, however, that cases 3 and 6 are worst

far from π_0 .

It must be stated clearly: The experimenter does not have a free choice of the values of k and b , but should judge what values of k and b reflect his prior knowledge of π_0 best.

4.3 Poisson distribution

Let $X_1, \dots, X_n$ be independent random variables from a distribution with probability mass function $f(x) = e^{-\lambda} \lambda^x / x!$, $x = 0, 1, 2, \dots$ and suppose that λ is a random variable with density function

$$g(\lambda) = \begin{cases} b a e^{-a\lambda} & \text{for } \lambda \geq 0 \text{ and } \lambda \neq \lambda_0 \\ 1 - b & \text{for } \lambda = \lambda_0. \end{cases}$$

The Bayes estimator for λ is given by

$$\hat{\lambda}_0 = \frac{ba\Gamma(r+2)/(n+a)^{r+2} + (1-b)e^{-n\lambda_0}\lambda_0^{r+1}}{ba\Gamma(r+1)/(n+a)^{r+1} + (1-b)e^{-n\lambda_0}\lambda_0^r}$$

with $r = \sum_{i=1}^n x_i$. If $b = 1$ $\hat{\lambda}_0$ reduces to the ordinary Bayes estimator $(r+1)/(n+a)$.

For $T_B = k r/n + (1-k)\hat{\lambda}_0$ we calculate $MSE(T_B)$ as in (3.10).

Looking at a typical case ($\eta_0 = 10$, $a = 1$) we see in figure 4 that for k and especially b small we have high REF around $\eta = \eta_0$ (curves 1 and 3 have maxima 14.8 and 16 respectively). For k and b large REF is not very high around η_0 , but has the slight advantage that it does not decline too far below 1 outside the effective interval. Case 6 : $k = 0.5$; $b = 0$ has the longest effective interval and still has a reasonable REF around η_0 .

4.4 Normal distribution

Let $X_1, \dots, X_n$ be independent random variables from a normal (μ, σ^2) distribution and let μ have a distribution

$$g(\mu) = \begin{cases} \frac{1}{(b/\sigma_0(2\pi)^{1/2})} \exp[-(\mu-\mu_0)^2/2\sigma_0^2] & \text{for } \mu \neq \mu_0 \\ 1-b & \text{for } \mu = \mu_0. \end{cases}$$

The Bayes estimator for μ is given by

$$\hat{\mu}_0 = (1-b)\mu_0 e^u / [\sigma^n (2\pi)^{n/2} f(x)] + b v e^w / [\sigma^{n-1} (2\pi)^{n/2} (\sigma^2 + n\sigma_0^2) f(x)]$$

where $u = -\sum_i (x_i - \mu_0)^2 / 2\sigma_0^2$, $v = (\mu_0^2 \sigma_0^2 + n\bar{x}\sigma_0^2) / (\sigma_0^2 + n\sigma_0^2)^{\frac{1}{2}}$,

$$w = (v^2 - \mu_0^2 \sigma_0^2 - \sigma_0^2 \sum_i x_i^2) / (2\sigma_0^2 \sigma_0^2)$$

and

$$f(x) = b e^w / [\sigma_0^{n-1} (2\pi)^{\frac{n}{2}} (\sigma_0^2 + n\sigma_0^2)^{\frac{1}{2}}] + (1-b) e^u / [\sigma_0^n (2\pi)^{\frac{n}{2}}].$$

Now for $T_B = k\bar{x} + (1-k)\hat{\mu}_0$, $MSE(T_B)$ is similar to (3.5).

In figure 5 we have $REF(T_B, \bar{x})$ for $\sigma_0^2 = 1$. Incidentally, the smaller σ_0^2 , the larger the relative efficiencies in $\delta = 0$ and the shorter the effective intervals. The conclusions regarding k and b are similar to those of the binomial and Poisson distributions. Curves 1 and 3 have maxima 8,8 and 16 respectively.

4.5 Conclusions

In practice k and b are parameters whose values depend on the experimenter's a priori knowledge of the unknown parameter θ and its prior distribution. Imagine that the buyer of a certain model car knows from road tests published by motoring journals that the average petrol consumption at a certain speed is approximately μ_0 . He may feel that it is appropriate to use a normal $(\mu_0, 1)$ distribution of μ around μ_0 and to place a weight $(1-b) = 0,50$ in μ_0 . Feeling quite confident about the choice of prior distribution, he may judge that $(1-k) = 0,75$ approximately reflects this confidence. With $k = 0,25$ and $b = 0,50$ his REF curve is number 5 in figure 5.

High relative efficiencies around θ_0 are obtained in the case $b = 0$, that is when our estimator T_B reduces to the ordinary shrinkage estimator T_L of Lemmer (1981). The maximum REF and the length of the effective interval then depend upon the value of k . The Bayesian shrinkage estimators ($0 < b < 1$) have an advantage over the ordinary shrinkage estimator T_L in the fact that their REF's outside the effective intervals do not decline as much as those of the ordinary shrinkage estimator. At the other end of the scale we have the ordinary Bayes estimator ($b = 1, k = 0$) which has a very long effective interval, but does not reach a high relative efficiency.

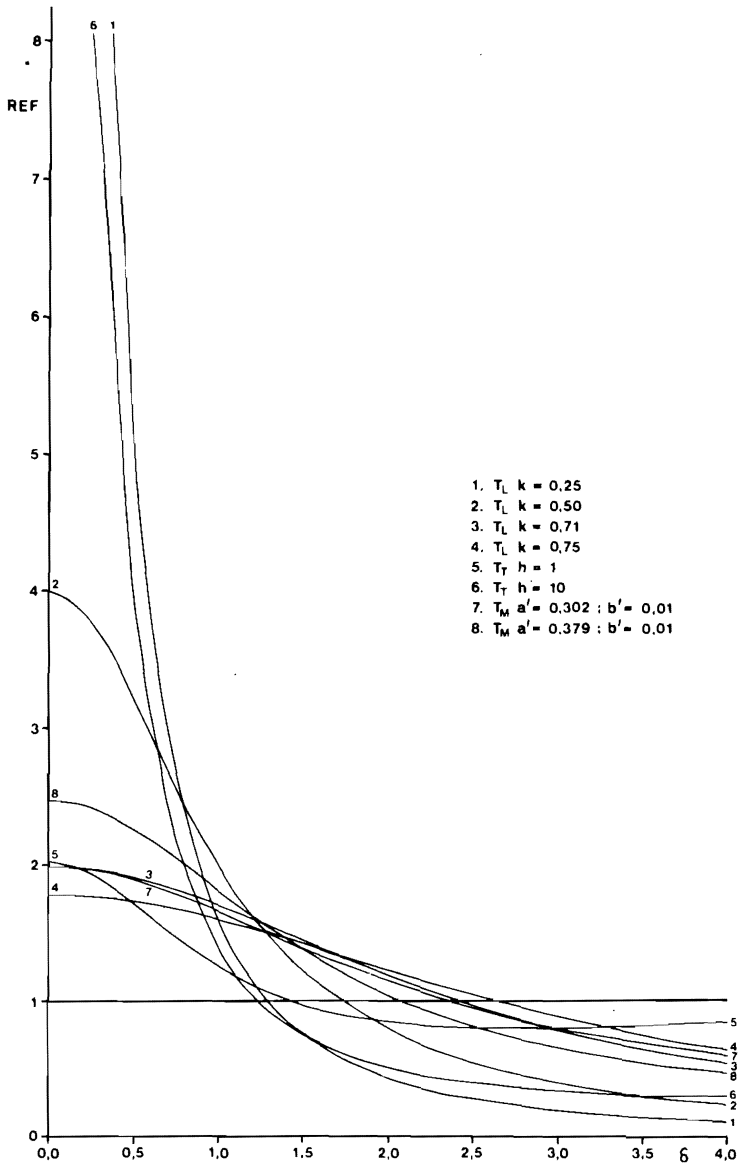


Figure 1 : Normal distribution

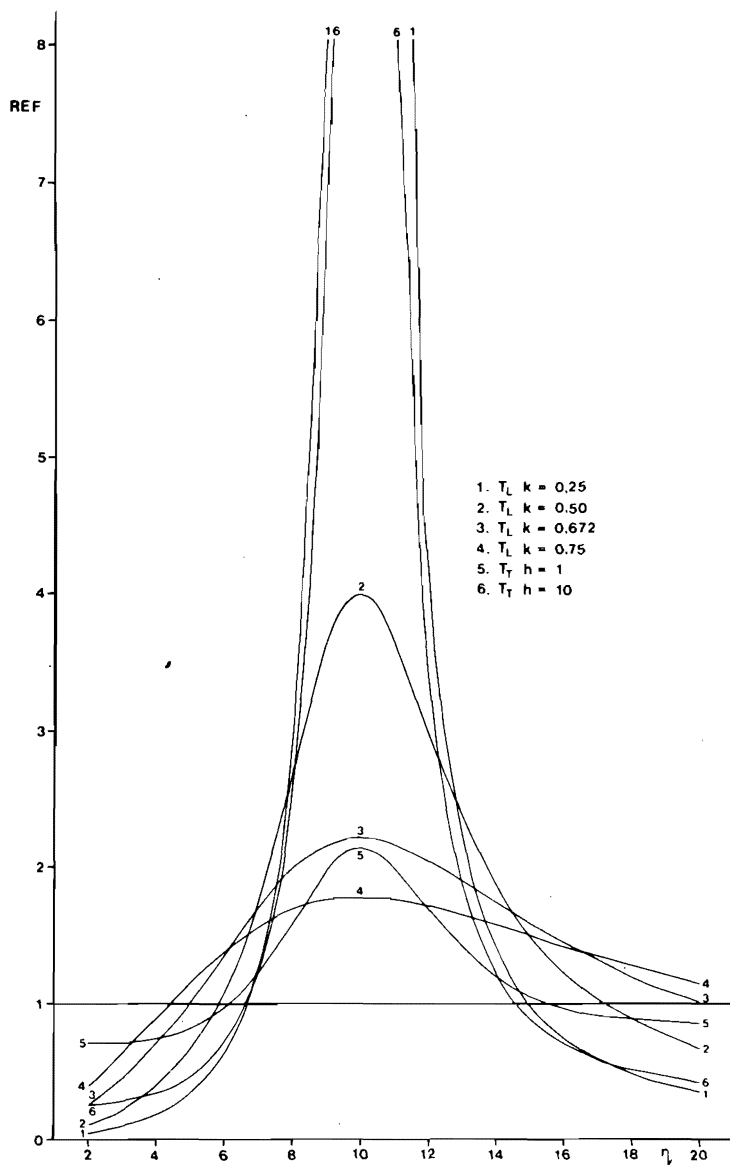


Figure 2 : Gamma distribution $\eta_0 = 10$

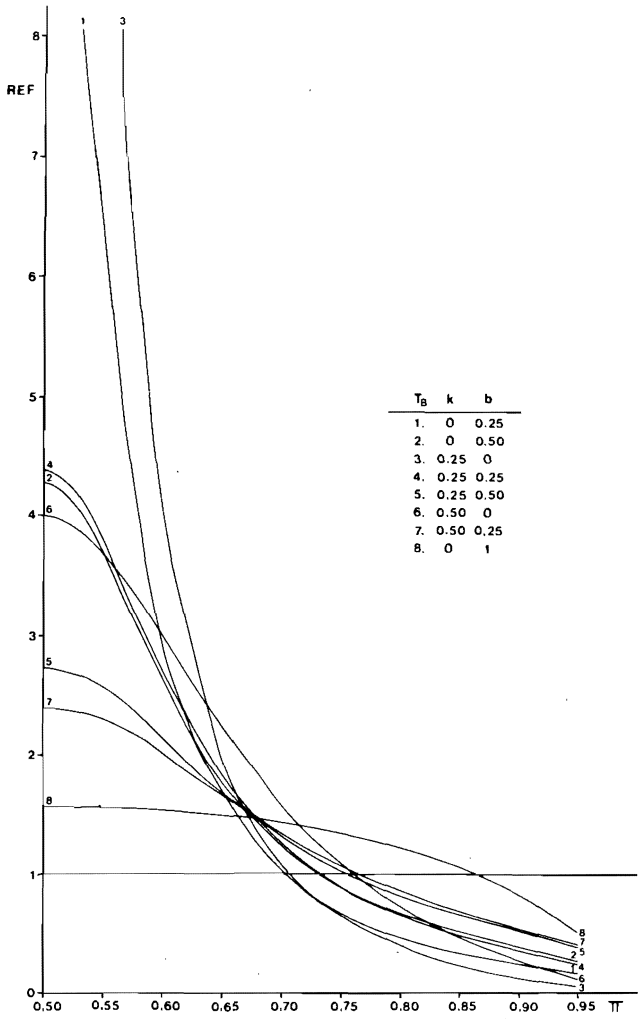


Figure 3 : Binomial distribution $n = 8, \pi_0 = 0,5$

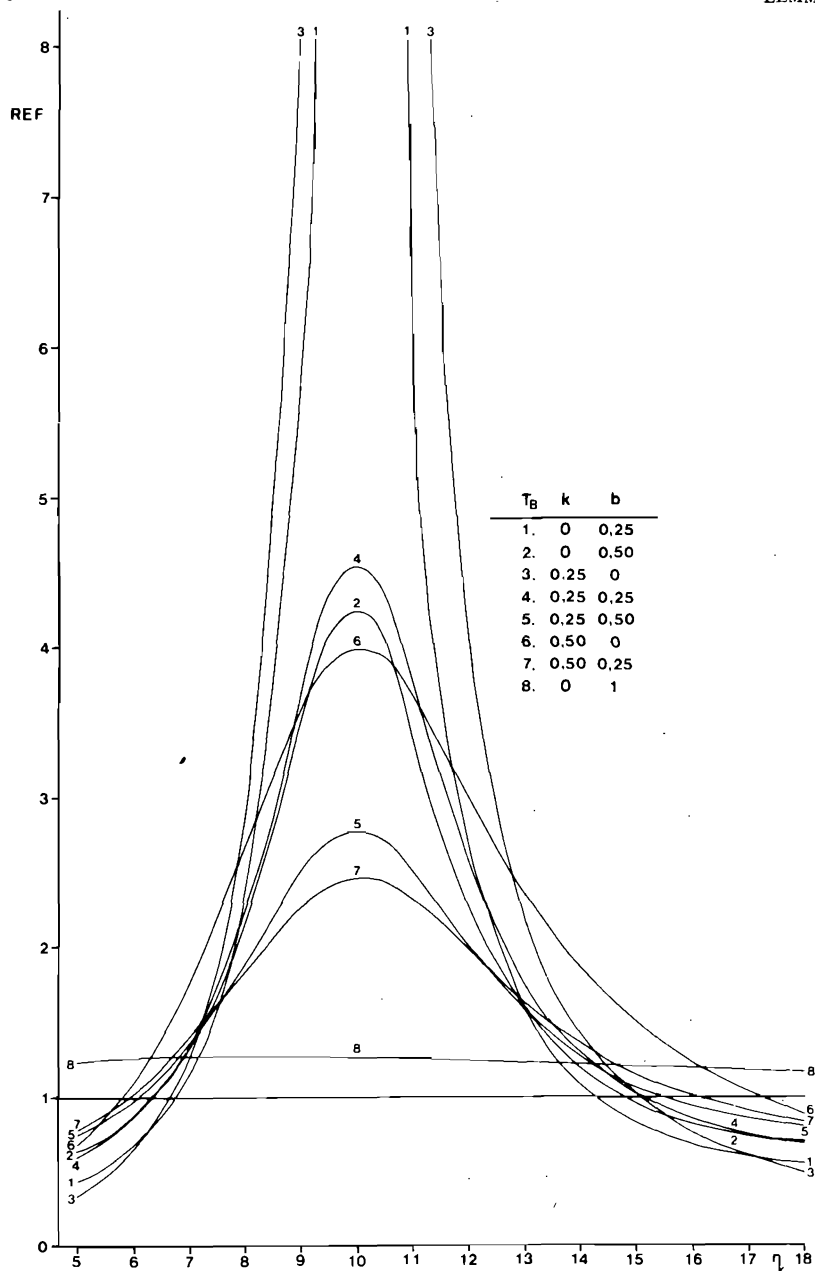


Figure 4: Poisson distribution $\eta_0 = 10$

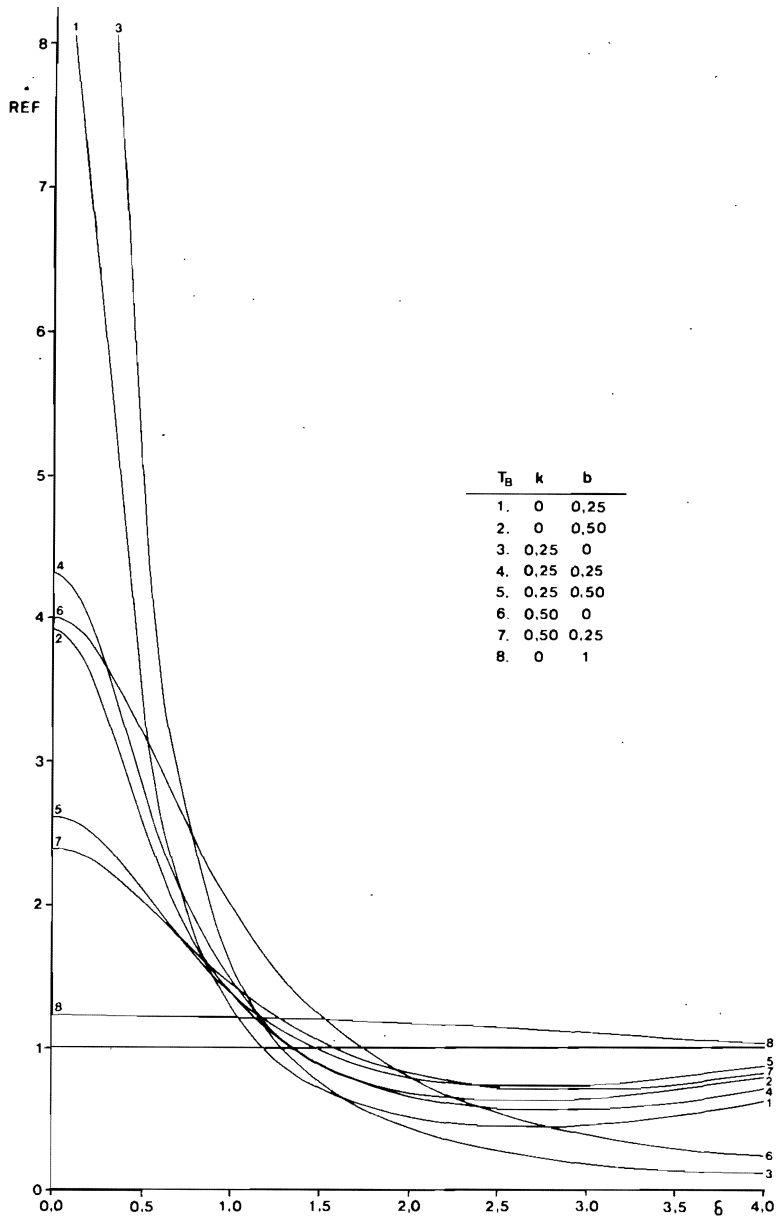


Figure 5: Normal distribution

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